SAPP: Student Academic Performance Predictor

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***Abstract*—Post COVID-19 pandemic, the education system has significantly shifted towards blended and hybrid learning approaches. In most cases, course delivery and materials are now being hosted online through various Learning Management Systems (LMS). In the recent years, researchers in the education paradigm have proposed methodologies to find key performance indicators from the LMS data and used Machine Learning (ML) models to identify students who require an early intervention to improve their academic performance. Although these ML-based solutions show high accuracy, these solutions are often restricted to a single course/module belonging to a specific department or university. Moreover, they evaluate a fixed set of ML models to determine the best performer. To address this, we propose a dynamic Student Academic Performance Predictor (SAPP) tool which can work for different types of modules having diverse student records. The tool can predict poorly performing students to make early intervention and improve their academic performance. As proof of concept, the tool has been designed as a Python application which can take LMS data from different modules and predict a list of students requiring early intervention. The application uses the best performing ML model out of 18 ML models. Initial results using LMS data collected from two different modules running at a UK university give us early indication that the SAPP tool can accurately predict student performance for various modules with diverse range of students.**

Keywords— student performance, prediction, machine learning, higher education

I. Introduction

We have seen a large growth in the use of online Learning Management Systems (LMS) in the past few years [1]. This has brought many benefits for all the stakeholders in an educational institute, including students, teachers, support staff, and the institute itself. One of the key benefits is the availability of students' daily records for each module in a course provided by LMS, for example, the Canvas LMS includes student analytics which present students' participation in a course including activities such as discussions and formative assignments, the contents they have viewed, submission details, and performance in weekly tasks/tests, etc. [2]. Additionally, for most computer science courses which use remote cloud-based servers for practical labs, there may be additional data such as log-in times, etc. The prevalence, presentation and near real-time nature of the data available through digital learning platforms enable the early intervention by the teaching team to promote engagement and improve academic outcomes [3-5].

In recent years, researchers [6-16] have proposed various methodologies and Machine Learning (ML) models to identify and predict low performing students who require an early intervention. In most cases, these ML-based solutions show high accuracy (>90%) in predicting student performance. However, these solutions are tailored for specific modules with a specific set of performance indicators generated by fixed LMS. Also, they work with only a handful set of ML models, without exploring the vast set of ML models that are available these days.

If we take two examples of Module-A (5 features, 50 students) and Module-B (10 features, 200 students); an ML model, Model-X may not perform correctly enough for Module-A, but it may perform very accurately for Module-B. In such a scenario, we need a different ML model, maybe Model-Y or Model-Z, which can provide us satisfactory results for Module-A. In an ideal world, we would like to have a solution which can accurately predict student performance for both Module-A and Module-B. In this paper, we propose the SAPP (Student Academic Performance Predictor) tool to provide such a dynamic solution.

The SAPP tool can take data for various modules generated by different educational institutes and then use a range of ML models to choose the best one to predict student performance. As proof of concept, the tool has been designed as a Python application which can take LMS data from different modules and predict a list of students requiring early intervention. The application provides 18 ML models to select the best performing model. Initial results using LMS data collected from two different modules running at a UK university give us early indication that the SAPP tool can accurately predict student performance for various modules with diverse range of students.

The paper makes the following key contributions:

- We present the SAPP tool as a web application and propose its architecture and workflow.
- We provide the SAPP prototype as a proof of concept for the SAPP web application.
- We use 18 ML models to evaluate the performance of the SAPP prototype.

The remainder of the paper is organised as follows. Section II presents related work in student performance prediction. Section III discusses the proposed SAPP tool, its architecture, workflow, web application, and prototype. Preliminary experimental results are evaluated and discussed in Section IV. Section V concludes the paper and discusses future work.

II. RELATED WORK

With the advancement of data science and ML, researchers working on education system development have come up with so many ways to improve student academic performance by applying various ML models.

Sangala et al. [6] used Logistic Regression on student data collected from a computer science faculty in an Indonesian private university. Experimental results show that Logistic Regression can achieve an accuracy of 91% in predicting student performance. Karmagatri et al. [7] applied a Decision Tree algorithm to predict students' likelihood to pass, which helps in interpreting the results more intuitively, and thereby assists in making informed decisions. Remegio [8] implemented Linear Regression, Decision Tree regression, Random Forest regression, and Neural Network regression to evaluate their performance and decide the best model to predict student performance. They found Random Forest regression to be performing the best with an R-squared score of 0.961.

Zhang et al. [9] ran experiments on blended learning data collected from the "Fundamentals of Programming (C)" course at Qinghai University from 2016 to 2019. They employed four ML models (Support Vector Machine, Random Forest, XGBoost, and Gradient Boosting decision tree) to evaluate student performance. The experimental results revealed that Support Vector Machine model is the best performer with an accuracy of 95%. Widyaningsih [10] used a semi-supervised learning approach (K-Means Clustering and the Naive Bayes Classifier) to classify the performance of first-year students in the department of mathematics. This approach achieved an accuracy of 96%. Ali et al [11] applied several machine learning classifiers such as Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), Support Vector Machine (SVM), and Naive Bayes (NB), to predict student performance. They also used a hybrid model to reduce data dimensionality and improve overall performance. The results show that CNN-RNN outperforms the other methods with an accuracy of 89.1%.

The work in [12] provides a comparison of supervised ML techniques in predicting student performance, specifically to identify the students with a "high risk" of dropping out from the course. Out of all the techniques, the highest precision was obtained by artificial neural networks. Similarly, in [13] researchers compared the performance of a number of supervised ML algorithms, such as Decision Tree, Naïve Bayes, Logistic Regression, Support Vector Machine, K-Nearest Neighbor, Sequential Minimal Optimisation, and Neural Network. They used the data collected from courses in the bachelor's study programmes in the University of Basra, for academic years 2017–2018 and 2018–2019. Experimental results show that Logistic Regression classifier is the most accurate in predicting student performance (68.7% for passed and 88.8% for failed).

Researchers in [14] proposed a new Performance Factors Analysis (PFA) approach based on different ML models such as Random Forest, AdaBoost, and XGBoost to increase the accuracy of student performance prediction. They evaluated the models using three different datasets. The evaluation results reveal that the scalable XGBoost outperforms the other models. In [15] researchers used various ML models such as Decision Tree, Naive Bayes, Logistic Regression, Multilayer Perceptron (MLP), Neural Network, and Support Vector Machine in order to detect at-risk, fail and excellent students in the early stages of a course. Evaluation results show the best accuracy for MLP. The research in [16] collected samples from 162,030 students from private and public universities in Colombia. They demonstrated that artificial neural networks could outperform other ML algorithms to classify between high and low performing students.

There are some earlier works in this paradigm as well. He et al. [17] proposed two transfer learning algorithms, namely "Sequentially Smoothed Logistic Regression (LR-SEQ) and Simultaneously Smoothed Logistic Regression (LR-SIM)". Comparing the results with the baseline Logistic Regression (LR) algorithm, they found that LR-SIM outperformed the LR-SEQ in terms of AUC. This research shows promising results in doing prediction at the early stage of admission. Kotsiaritis et al. [18] proposed a combinational incremental ensemble technique for student performance prediction. They combined three classifiers each of which calculated the prediction output. They then used a voting method to select the overall final prediction.

We can see that researchers have proposed many highly accurate ML based algorithms to predict student academic performance. We can also observe that there is almost always a new ML model that is coming out as the best performer when running on a new dataset collected from a new course/university. That means there is no single winner, no single ML solution that can work for all kinds of student data. This is where the SAPP tool is going to help as a dynamic tool which can decide on the fly which ML model is going to be the best for the dataset that it is fed with and use that model to predict student performance.

III. PROPOSED SAPP TOOL

In this section, we present the SAPP tool and discuss its architecture, workflow, web application, and prototype.

A. SAPP Architecture

SAPP tool is designed to be dynamic, which can take input for a variety of modules from any LMS source and run a number of ML models to enhance the accuracy of predicting student performance. Fig. 1 presents the architecture of the tool. The tool consists of a number of components which are discussed as follows:

1) GUI: The GUI is the interface using which a user can interact with SAPP. The GUI allows the users to submit their LMS data along with the options to predict the performance in the middle of the term time or after the term. The GUI also provides the option to select whether the user wants to save the prediction results for future use or whether the user wants to retrieve previously saved results. To allow the users to retrieve their saved results, the GUI provides a user registration and login mechanism.

The GUI generates error message if the data is not in the correct format or if there are not enough records or features to run the ML models. If there is no error, the GUI runs the ML models on the provided LMS data and generates a report which will include a list of the low performing students which may require early intervention.

2) Pre-processor: The pre-processor cleans the raw data that is provided by the users via the GUI and makes it ready for running ML models on them.

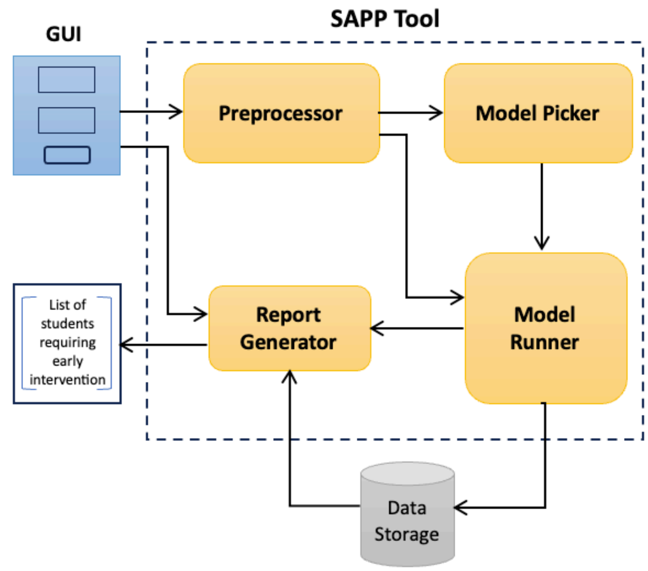

Fig. 1. SAPP Tool Architecture

This cleaning includes filtering out rows with null/empty values, removing columns that are irrelevant, renaming some of the columns to adjust for ML models, transforming non-numerical labels to numerical labels. This also includes techniques like SMOTE (Synthetic Minority Oversampling Technique) [14] to resolve class imbalance problems when using classification models.

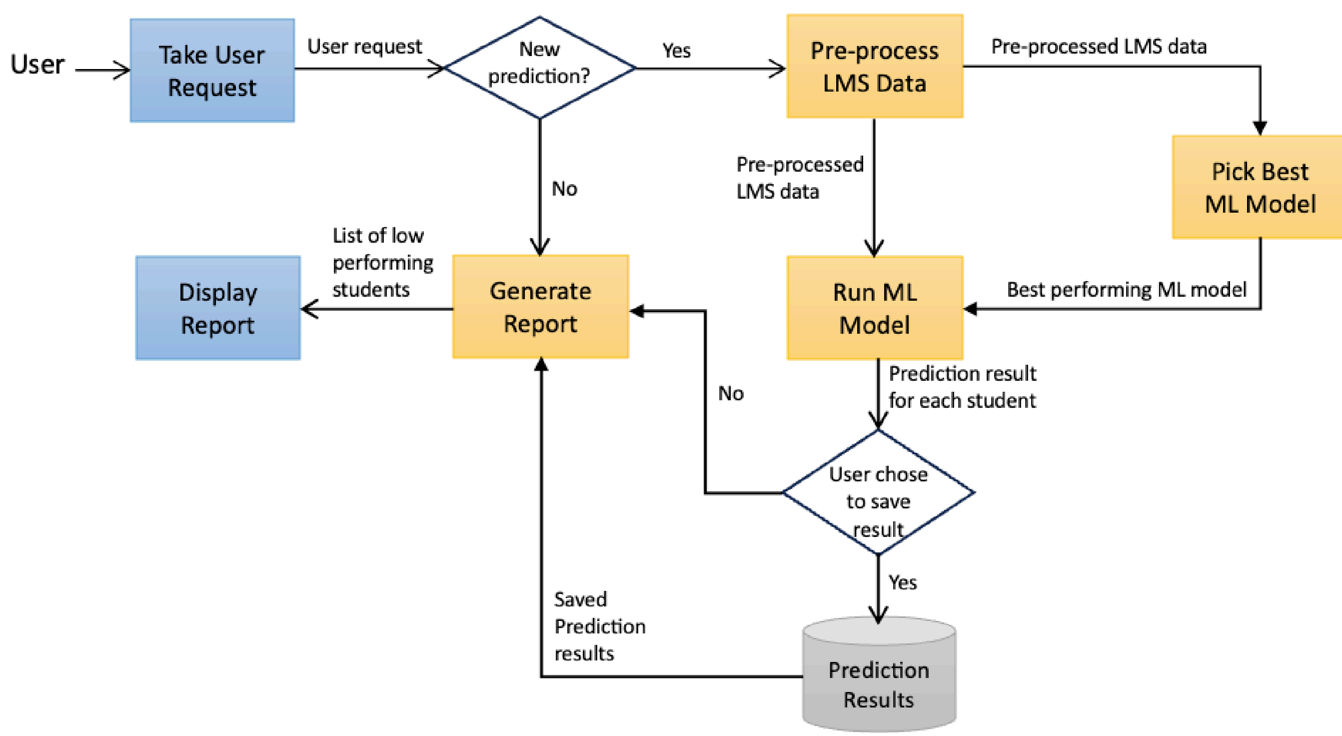

Fig. 2. SAPP Workflow

3) Model Picker: This is the most important component of SAPP, which runs a set of ML models on the pre-processed data and selects the best performing model (in terms of accuracy) to use for predicting the student performance. The Model Picker evaluates the performance of the ML models by splitting the data into training and testing dataset. The Model Picker uses two kinds of pre-processed data: one before SMOTE is applied and one after SMOTE is applied. This is to determine if using SMOTE would improve the performance or not, and then to provide this recommendation to Model Runner.

4) Model Runner: This is where the best performing ML model is run on the pre-processed data to predict student performance. The prediction results are sent to the Report Generator to prepare the report for the user. If the user selects the option to save the results for future use, the results are stored in a database.

5) Report Generator: This component generates the report to be displayed on the GUI, ideally a list of students who may require early intervention based on their predicted academic performance. The Report Generator can be accessed directly by the GUI if a user wants to check a previously generated prediction result.

B. *SAPP Workflow*

Fig. 2 depicts how different components in the SAPP tool work together right from taking user request to displaying the report on list of low performing students.

C. *SAPP Web Application*

We envisage SAPP as a web application as depicted in Fig. 3, which can be hosted in a public cloud platform. The application includes four key components:

- **SAPP Web App:** This is the GUI through which users can interact with SAPP, submit their requests, and receive a report on student performance.
- **SAPP Runner:** This acts as the communicator between the user and the SAPP tool; this will receive the user requests/input data and perform some request/data validations before passing on the requests and the data to SAPP Engine.
- **SAPP Engine:** This hosts all the key components of the SAPP tool: pre-processor, model picker, model runner, and report generator.
- **Data Storage:** This is where all the reports are stored if students decide to save student performance reports for future use. This also stores user login details to provide the option to retrieve saved reports.

All these components can be hosted in Virtual Machines in a public/private cloud, and the web application can be made available for all the higher education institutes.

It is important to note here that we do not intend to present the fully working SAPP web application in this paper, rather we present a prototype of the application as proof of concept.

D. *SAPP Prototype*

We built a prototype of the SAPP web application in order to validate the concept and check the accuracy of prediction. The prototype can take LMS data as a CSV file and generate a list of students who are predicted to fail in the module. Fig. 4 showcases how the prototype works. To mimic the future real-world SAPP web application, the prototype implements a Python script as SAPP Runner and a Python module as SAPP Engine.

The Python script reads the LMS data (as a CSV file) and uses the SAPP Engine to predict student performance. The SAPP Engine performs the key steps of the SAPP workflow: pre-processing the raw LMS data, picking the best ML model, and running the best ML model to list the students who are predicted to fail in the module.

The SAPP Engine chooses the best performing model out of a list of 18 ML classification models (see Table I for the list). These models are selected based on our literature review [6-16], where researchers have found these models to be performing well for various modules on a range of students. To evaluate the performance of each of the models, the Engine splits the pre-processed data into 80:20 ratio (80% for training, and 20% for testing). This split ratio is decided after reviewing several evaluation techniques proposed by various researchers [6-11]. To resolve the issue of class imbalance, especially when the student data is too small, the Engine applies SMOTE. Researchers in [9] has found SMOTE to improve the accuracy of prediction when dataset is small.

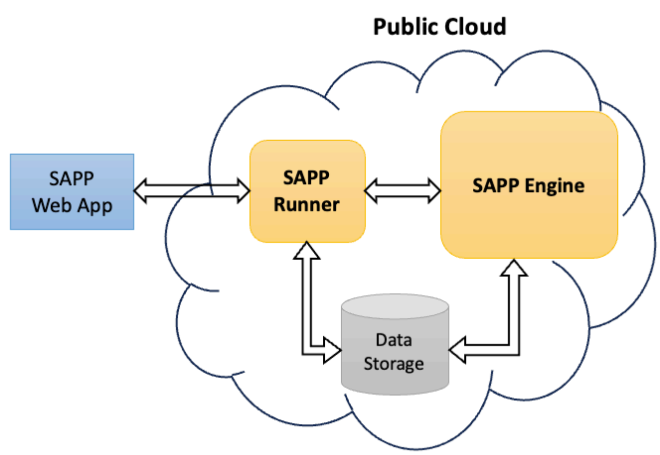

Fig. 3. SAPP Web Application

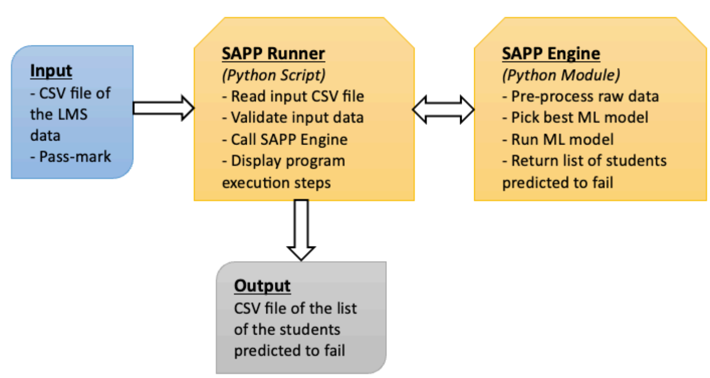

Fig. 4. SAPP Prototype

TABLE I. LIST OF ML MODELS

Model	Model Name
LOG_REG	Logistic Regression
KNN	K-Nearest Neighbors
SVM	Support Vector Machine
DT	Decision Tree
RF	Random Forest
GBM	Gradient Boosting Machine
ADA	AdaBoost
XGB	XGBoost
LGBM	LightGBM
NB	Naive Bayes
MLP	Multi-Layer Perceptron
LDA	Linear Discriminant Analysis
QDA	Quadratic Discriminant Analysis
BAG	Bagging
ET	Extra Trees
HGBOOST	HistGradient Boosting
RIDGE	Ridge Classifier
SGD	Stochastic Gradient Descent

IV. EXPERIMENTAL RESULTS

To test the performance of SAPP prototype, we collected student data from two different modules running in the Computer Science department in a UK university. The data is generated by Canvas, and it includes a number of performance metrics. Among those metrics, three key performance indicators stand out, which are pageviews, participation, and total marks. Pageviews counts the number of times a student has viewed the Canvas pages which include module contents. Participation counts the number of times a student has taken part in the Canvas activities. These metrics were chosen based on our previous work [20], which demonstrates how these metrics correlate with students' total marks. Both the modules have the same performance metrics, but the number of students differs. The first module, from Stage 3, is Cloud Computing (we refer to this as Module 1) with 236 students; while the second module, from Stage 1, is Software Design Principles (we refer to this as Module 2) with 40 students.

As the SAPP prototype is designed to list the students who are predicted to fail the module, in addition to the performance metrics, we needed to provide a pass mark. For Module 1 we provided the pass mark as 40 and for Module 2 we provided the pass mark as 50. Once the student data and pass marks are supplied to the SAPP prototype, the SAPP Engine first evaluates the performance of the 18 ML classification models that are included in it and then selects the best performing model (in terms of accuracy) to predict the students who are going to fail the module.

TABLE II. PERFORMANCE ON MODULE 1

Model	Accuracy%	Precision%	Recall%	F1_Score%
LOG_REG	89.58	89.13	100.00	94.25
KNN	85.42	85.42	100.00	92.14
SVM	85.42	85.42	100.00	92.14
DT	93.75	100.00	92.68	96.20
RF	91.67	91.11	100.00	95.35
GBM	**95.83**	**97.56**	**97.56**	**97.56**
ADA	93.75	93.18	100.00	96.47
XGB	93.75	93.18	100.00	96.47
LightGBM	85.42	86.96	97.56	91.96
NB	85.42	85.42	100.00	92.14
MLP	85.42	85.42	100.00	92.14
LDA	85.42	85.42	100.00	92.14
QDA	85.42	85.42	100.00	92.14
BAG	**95.83**	**97.56**	**97.56**	**97.56**
ET	91.67	91.11	100.00	95.35
HGBOOST	85.42	88.64	95.12	91.77
RIDGE	85.42	85.42	100.00	92.14
SGD	85.42	85.42	100.00	92.14

TABLE III. PERFORMANCE ON MODULE 2

Model	Accuracy%	Precision%	Recall%	F1_Score%
LOG_REG	28.57	33.33	66.67	44.44
KNN	28.57	33.33	66.67	44.44
SVM	28.57	33.33	66.67	44.44
DT	42.86	42.86	100.00	60.00
RF	42.86	42.86	100.00	60.00
GBM	42.86	42.86	100.00	60.00
ADA	42.86	42.86	100.00	60.00
XGB	28.57	33.33	66.67	44.44
LightGBM	42.86	42.86	100.00	60.00
NB	28.57	33.33	66.67	44.44
MLP	14.29	20.00	33.33	25.00
LDA	28.57	33.33	66.67	44.44
QDA	**57.14**	**50.00**	**66.67**	**57.14**
BAG	28.57	33.33	66.67	44.44
ET	42.86	42.86	100.00	60.00
HGBOOST	42.86	42.86	100.00	60.00
RIDGE	28.57	33.33	66.67	44.44
SGD	42.86	42.86	100.00	60.00

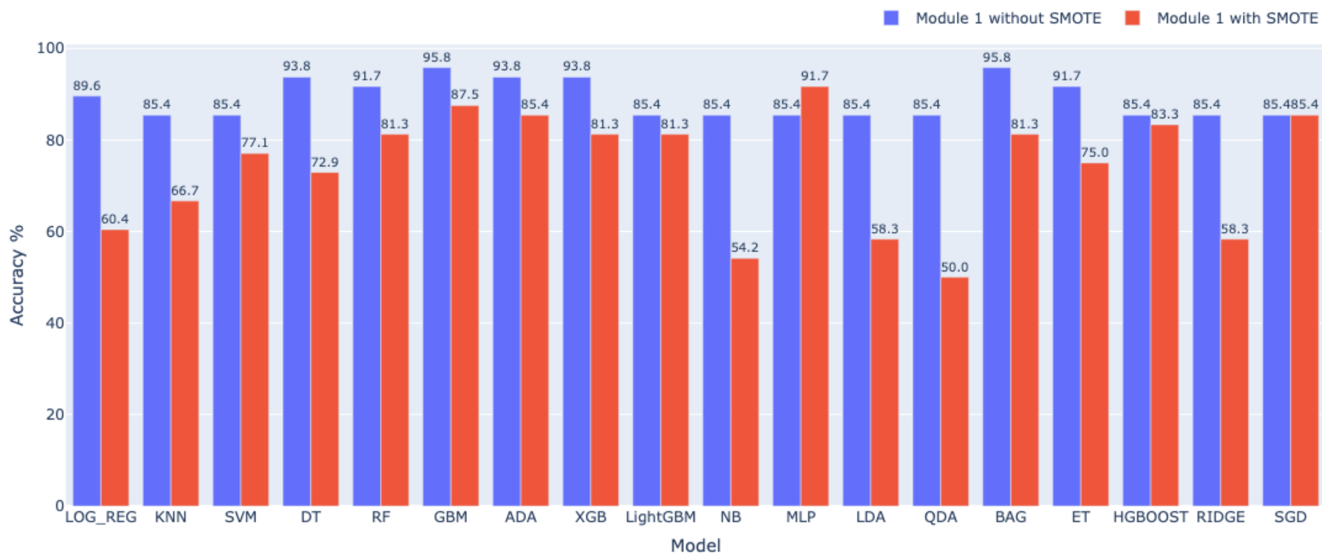

Fig. 5. Accuracy for Module 1: with SMOTE vs without SMOTE

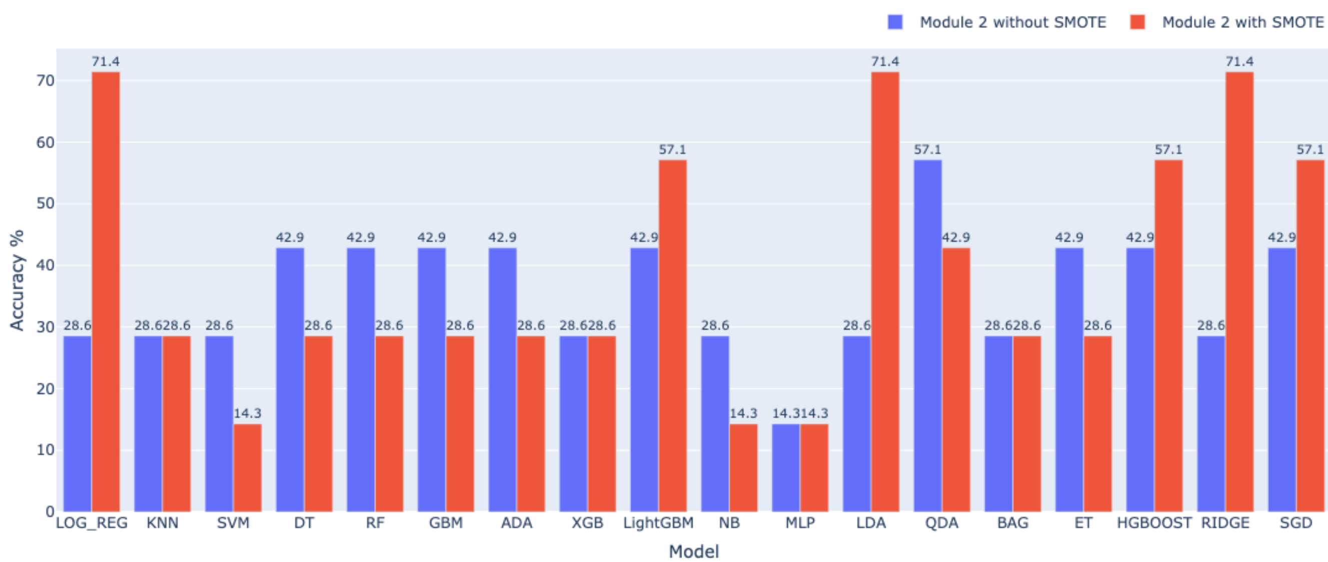

Fig. 6. Accuracy for Module 2: with SMOTE vs without SMOTE

To discuss the performance of the SAPP prototype, we use four different performance metrics:

- **Accuracy** is the ratio of correctly predicted passes and fails out of all the students.
- **Precision** is the measure of correctly identified fails out of all failed students.
- **Recall** is the measure of correctly identified fails over all the students in the data.
- **F1 Score** is the measure of combining both the precision and the recall; it is measured as the harmonic mean of the two.

Table II and III present the performance metrics of the models for Module 1 and Module 2, respectively. From the tables, we can see that GBM (Gradient Boosting Machine), and BAG (Bagging) are the best performing models for Module 1 with an Accuracy of 95.83%, and its precision, recall and F1 score values are high as well. Whereas for Module 2, QDA (Quadratic Discriminant Analysis) is the best performing model with an Accuracy of only 57.14%. Clearly, the models didn't work well for Module 2. This is most likely due to the fact that Module 2 has a much smaller dataset (only 40 students) as opposed to Module 1 (236 students). To resolve this issue, we applied SMOTE in the SAPP Engine.

Fig. 5 and Fig. 6 present the bar plots of the comparative accuracy performance between when SMOTE was applied (with SMOTE) and when SMOTE was not applied (without SMOTE). From the plots we can observe that different ML models perform differently for the two modules, and SMOTE improves the performance for Module 2 (best performance rises from 57.14% to 71.4%), but drops the performance for Module 1 (best performance drops from 95.8% to 91.7%). This clearly indicates the importance of having different ML models and different pre-processing techniques (e.g. SMOTE) for different datasets, hence for different modules or for different year for the same module. SAPP provides this flexibility to use the best performing model and pre-processing techniques after evaluating the performance of a series of top ML models, making the tool fit for different modules with varying student numbers.

V. Conclusion and Future Work

In this paper, we have proposed a dynamic Student Academic Performance Predictor (SAPP) tool which can predict poorly performing students in order to make early intervention to improve their performance. The tool is dynamic in the sense that it can work for different types of modules having diverse student records. We have presented the architecture of the tool, its workflow, and a prototype as proof of concept. Evaluation results using 18 ML models on two different course modules validate the feasibility and high performance of the SAPP tool. In the future, we plan to deploy the tool as a web application, which can be accessed by a tutor in any higher education institute to predict student performance and take necessary actions. We will also extend the list of our ML models to include more classification and clustering models, and we will evaluate the tool using student data collected from a greater number of diversified modules (with variety of features, and student numbers) running at different higher education institutes.

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